SAFE & RESPONSIBLE AI SYSTEMS

Note: No Week 9 lecture slides were in the project — these notes are reconstructed from the assignment questions + accepted answers and standard course material. Every graded question is covered.

1 Regulating AI — Vertical vs Horizontal

Vertical regulation — applies **existing sector-specific laws** to AI (health, finance, autos each regulate AI within their own domain). Narrow, per-industry, reuses what's already on the books.
Horizontal regulation — creates **broad, new, risk-based laws across ALL sectors** (e.g. EU AI Act). One framework covering every use of AI, classified by risk.
Trade-off: vertical is faster & fits domain nuance but leaves gaps; horizontal is comprehensive but slow & blunt.

Three regulatory postures:

Posture	Who sets rules
Self-regulation	Industry alone; no govt enforcement
Co-regulation	Industry sets policy, **govt formally recognizes & enforces**
Strict/State regulation	Govt dictates rules directly

Upstream vs downstream (Q11): regulate **developers** (upstream) not **users** (downstream) — users **face high market pressure to adopt AI** and can't opt out, so the duty belongs with those who build it.

Assignment: Vertical = existing sector-specific laws; Horizontal = broad new risk-based laws across all sectors (Q1). Co-regulation = industry policies formally recognized & enforced by govt (Q5). Upstream regulation better → users face high market pressure to adopt (Q11).

2 Guardrails — maturity differs

Responsible-AI guardrails vary in mathematical maturity:
- **Data privacy** — **well-understood**: **formally bounded & guaranteed** (differential privacy gives provable ε-guarantees; see Weeks 6–7)
- **Model explainability** — weak: no exact bounds, hard to guarantee or even define

Lesson: prefer guardrails you can **prove**, and treat the fuzzy ones (explainability, fairness) as best-effort.

Privacy-preserving training (Q4):

- **Differential Privacy** — calibrated noise, provable privacy
- **Federated Learning** — train across devices, raw data never centralised

Distractors: **Model Quantization** (compression / efficiency) and **Data Augmentation** (robustness) are NOT privacy techniques.

Assignment: privacy is better-understood because it is **formally bounded & guaranteed mathematically** (Q2). Privacy-preserving training = **Differential Privacy + Federated Learning** (Q4).

3 Autonomy, Copyright, Energy & Jailbreaks

Autonomy & human agency (Q3): as AI becomes autonomous agents, the main trade-off is that **individual human agency decreases as decision-making is delegated to AI.**
Copyright inside a model (Q6): to remove copyrighted data, **the model may need retraining or specialized unlearning** to erase the data's influence (ties to Week 3).

Back Translation Defense (Q9): anti-jailbreak — take an unmoderated output, **reverse-infer the prompt that produced it, and run that prompt through safety filters.** Unsafe inferred prompt → block output.
Green AI / energy (Q8): cut LLM energy & carbon by **moving inference to CPUs/Edge devices instead of GPUs** for simple queries; don't send trivial queries to huge GPU clusters.

4 Responsible AI (RAI) — principles, budget, fairness tree

RAI principles (Q10): Fairness, Explainability, Accountability, Privacy. NOT 'AI Autonomy', NOT 'Fast Deployment'.
Safety budget (Q7): spend about **10% of development budget on safety & responsible-AI research.**
Systems approach (Q12): a loop — analyze root causes → **measure & detect bias** → mitigate via interventions → monitor & evaluate → all to **improve fairness of AI systems.**

Fairness Tree Framework (Q11): choose a fairness metric from two questions — is the intervention **punitive vs assistive**, and which group do you care about?

Situation	Metric
Punitive + predictive equity among those intervened on	**FDR Parity**
Punitive + among those who need it	FPR / FN parity
Assistive	equal selection / recall parity

Punitive intervention (e.g. arrest, denial) → you care about **false discoveries** → **FDR Parity.**

Assignment quick-key: Q1 vertical=sector/horizontal=broad; Q2 privacy formally bounded; Q3 human agency decreases; Q4 DP+Federated; Q5 co-regulation; Q6 retrain/unlearn; Q7 10% safety budget; Q8 CPU/Edge inference; Q9 back-translation; Q10 Fairness/Explainability/Accountability/Privacy; Q11 FDR Parity; Q12 improve fairness of AI.